

Creating a One-Dimensional NumPy Array

Numbers from 1 to 10

Python | NumPy | Assignment 2

IN SIMPLE WORDS

What is a 1D NumPy array?

Think of a row of lockers.

Lockers sit in one straight line, numbered one after another. A 1D array is the same: one row of values, all of the same type, stored in order.

One row, one dimension.

1D

array of 10 values, one axis

example.py

```
import numpy as np

arr = np.arange(1, 11)
print(arr)
```

output

```
[ 1  2  3  4  5  6  7  8  9 10]
```

Using np.arange()

`np.arange(start, stop)` makes numbers from start up to, but not including, stop. So to get 1 to 10, the stop value is 11.

```
arange.py

import numpy as np

arr = np.arange(1, 11)
print(arr)
```

Output

```
output

[ 1  2  3  4  5  6  7  8  9 10]
```

Alternative: `np.array(range(1, 11))` gives the same result.

Check: is it one-dimensional?

ndim tells the number of dimensions, and shape tells the size of each one.

```
check.py

import numpy as np

arr = np.arange(1, 11)
print(arr.ndim)
print(arr.shape)
```

Output

```
output

1
(10,)
```

1 dimension, 10 elements

FULL PROGRAM

Code, explanation and output

1. Import NumPy as np.
2. np.arange(1, 11) creates the numbers 1 to 10.
3. The result is a 1D array stored in arr.
4. print() shows the array.

```
numbers.py

# Create a 1D array
import numpy as np

arr = np.arange(1, 11)

print(arr)
print(arr.ndim)
```

```
output

[ 1  2  3  4  5  6  7  8  9 10]
1
```